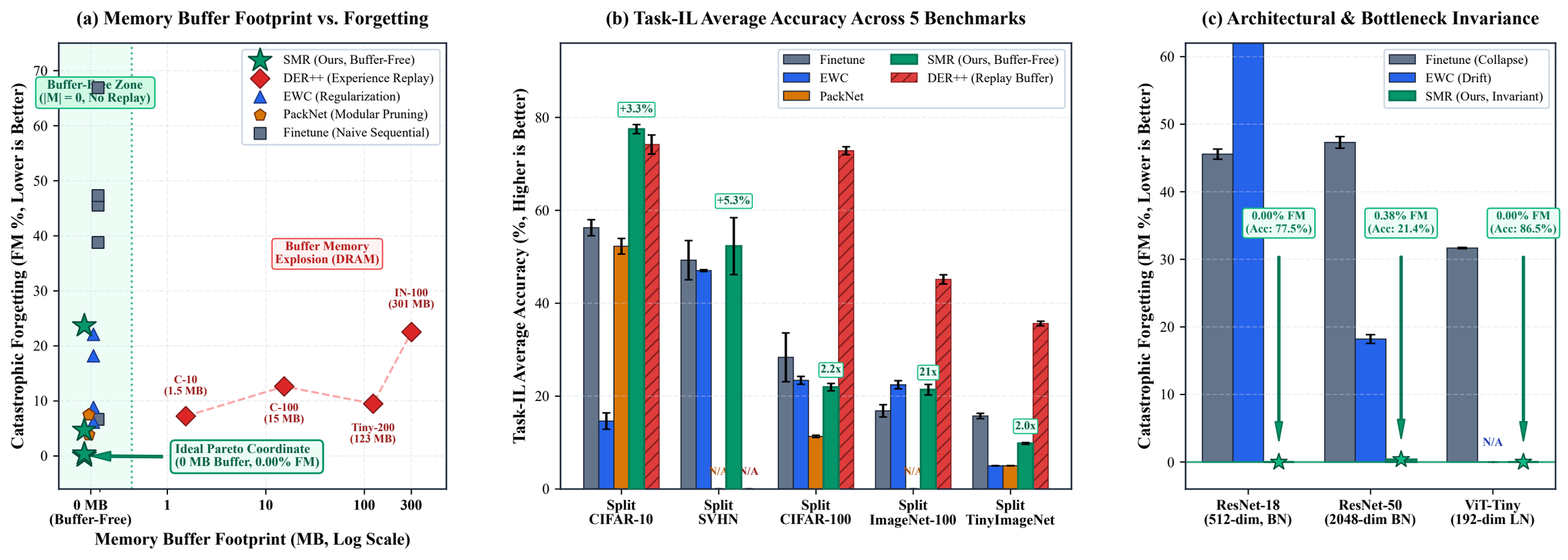

(a) Memory Buffer Footprint vs. Forgetting
SMR (Ours, Buffer-Free)
DER++ (Experience Replay)
EWC (Regularization)
PackNet (Modular Pruning)
Finetune (Naive Sequential)
Buffer-Free Zone
(|M| = 0, No Replay)
Buffer Memory
Explosion (DRAM)
C-10
(1.5 MB)
C-100
(15 MB)
Tiny-200
(123 MB)
IN-100
(301 MB)
Ideal Pareto Coordinate
(0 MB Buffer, 0.00% FM)
Catastrophic Forgetting (FM %, Lower is Better)
0 MB
(Buffer-Free)
1
10
100
300
Memory Buffer Footprint (MB, Log Scale)
(b) Task-IL Average Accuracy Across 5 Benchmarks
Finetune
EWC
PackNet
SMR (Ours, Buffer-Free)
DER++ (Replay Buffer)
+3.3%
+5.3%
2.2x
21x
2.0x
N/A
Task-IL Average Accuracy (%, Higher is Better)
Split
CIFAR-10
Split
SVHN
Split
CIFAR-100
Split
ImageNet-100
Split
TinyImageNet
(c) Architectural & Bottleneck Invariance
Finetune (Collapse)
EWC (Drift)
SMR (Ours, Invariant)
0.00% FM
(Acc: 77.5%)
0.38% FM
(Acc: 21.4%)
0.00% FM
(Acc: 86.5%)
N/A
Catastrophic Forgetting (FM %, Lower is Better)
ResNet-18
(512-dim, BN)
ResNet-50
(2048-dim BN)
ViT-Tiny
(192-dim LN)